UTOPI-AR: User Behavior Guided Token Pre-Compress for Efficient Egocentric Long-Video Understanding in AR Environment

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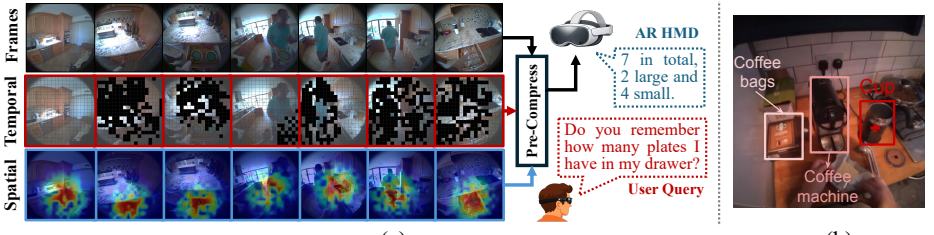
Abstract. As long-video understanding continues to gain traction, Augmented Reality (AR) devices emerge as a compelling deployment platform. However, egocentric videos exhibit significant temporal redundancy, imposing substantial memory and computational burdens on resource constrained AR hardware. To address this challenge, we introduce *UTOPI-AR*, a plug-and-play pre-compression module designed specifically for egocentric long videos on AR platforms. First, we leverage user motion signals to estimate inter-frame overlap and propose a viewpoint-shift-invariant pruning algorithm. Second, we exploit eye-tracking sensors to identify user fixations, enabling the separation of attended foreground content from background and its preferential retention via a fine-tuned neural network–based adjustment mechanism. Across multiple benchmarks and model backbones, *UTOPI-AR* achieves up to 95% compression while preserving performance.

Keywords: Video understanding · AR/VR · Vision-language model



(a) (b)

Fig. 1: (a) UTOPI-AR compresses temporal redundancy from viewpoint changes and spatial redundancy in user unattended regions. (b) An example illustrating how user attention varies with gaze location.

1 Introduction

Augmented Reality (AR) is emerging not only as a novel user interface but also as a powerful machine learning (ML) platform that integrates rich embodied experiences, including sensing, perception, memory, and speech. By tightly coupling the digital and physical worlds, AR enables real-time extraction of actionable information and context-aware decision making, driving transformative

advances across domains such as healthcare [16, 23, 42], education [6, 56, 59], and industry [17]. Beyond immersive applications, AR devices provide powerful platforms for ML by continuously generating rich multimodal sensory data. Equipped with cameras, eye-tracking, and motion sensors, they capture egocentric video along with real-time pose and behavioral signals. These heterogeneous, high-throughput streams create new opportunities for long-context ML models to learn correlations among multimodal signals, user attention, and environmental context.

As shown in Figure 1 (a), consider an AR head-mounted display (HMD) used during driving navigation. The device continuously captures and encodes multimodal inputs with ML models, and a brief user glance at a drawer must be reliably bound to the corresponding visual features. Later, answering a query such as "How many plates in my drawer?" requires an episodic memory module that can index temporal embeddings, retrieve the correct event, and interface with a downstream processing engine (e.g., a large model). This capability of *long video understanding* (LVU) transforms AR from a passive sensing platform into an intelligent personal assistant with long-context reasoning, marking a critical step toward next-generation AR applications. In education, students can revisit past demonstrations to reinforce learning; in industry, recorded tasks can be replayed to identify errors; in healthcare, surgeons can query moments when key anatomical landmarks appear; and in daily life, AR can serve as a memory aid for locating items such as keys. However, supporting LVU imposes severe constraints on resource-limited AR hardware. For example, a 100-second high-definition video sampled at 1 FPS, when encoded by a typical vision–language model (VLM), can require tens of gigabytes of memory, far exceeding the 8 to 16 GB capacity of current wearable devices [7, 45]. Lowering the sampling rate reduces memory usage but significantly harms representation quality and downstream accuracy. In addition, continuously running large encoder backbones for online encoding, retrieval, and reasoning is computationally prohibitive on AR devices. These challenges highlight the need for new algorithms that enable LVU with substantially improved efficiency.

To address the first challenge, we propose a *perception-guided approach* to improve the efficiency of LVU in AR, drawing inspiration from human attentional mechanisms. Human perception is inherently nonuniform, with the highest visual fidelity concentrated at the focus of attention and progressively decreasing with increasing eccentricity [9, 22, 54]. In egocentric AR video, most visual content is closely related to the user's ongoing task. Prior studies [11, 26, 63] show that human intent is strongly reflected in the questions users ask and the answers they seek. As a result, regions outside the focus of attention often contain substantial spatial redundancy and contribute little to downstream reasoning. Egocentric AR streams also exhibit strong temporal redundancy, as head and body movements are typically smooth, causing large overlap across consecutive frames (Figure 1 (a)). Our key idea is to exploit these spatial and temporal regularities by prioritizing computation and memory for behaviorally salient regions and events, reducing cost while preserving information needed for LVU.

Building on these observations, we introduce *UTOPI-AR*, a plug-and-play **U**ser behavior guided **TO**ken **P**re-compression module **I**n **AR** environments. UTOPI-AR improves efficiency by exploiting spatial redundancy in unattended regions and selecting important tokens using gaze-informed attention via a lightweight saliency gating head. To address temporal redundancy, we preserve full-fidelity representations of inter-frame overlap regions at their first appearance, estimated using AR motion signals, and replace them with compressed tokens in subsequent frames. Additionally, we also introduce *EgoEye*, a benchmark designed to study LVU in AR settings. EgoEye captures the **human-centric** and **attention-guided** nature of AR usage by aligning questions with users' gaze patterns. Overall, our contributions are summarized as follows:

- We introduce UTOPI-AR, a plug-and-play user behavior–guided token pre-compression module that leverages AR-native signals to compress visual tokens before the query stage. By jointly reducing spatial and temporal redundancies, it significantly lowers computational and memory costs.
- By integrating UTOPI-AR into multiple VLMs, we enable practical deployment on an AR simulation platform and unlock video understanding under tight resource constraints. We also introduce EgoEye, an egocentric video benchmark with diverse sensor signals, featuring both attention-driven and detail-oriented QA pairs.
- Across multiple egocentric LVU benchmark and diverse baseline model, our approach achieve consistent efficiency–accuracy benefits: maintaining baseline accuracy while achieving substantial token compression (use only 5% visual tokens on EgoSchema) and markedly reducing on-device memory cost.

2 Background and Related Work

2.1 Efficient Long Video Understanding

Contemporary vision–language models (VLMs), including proprietary systems [48, 57] and open source counterparts [10, 19, 67], transform images and videos into sequences of visual tokens and process them jointly with text in a unified framework. As a result, they support visual perception, reasoning, instruction following, and dialogue, making them a common backbone for video understanding [31, 36, 40, 55, 68]. However, most VLMs are typically large and computationally expensive, with substantial memory overhead. A significant portion of this overhead arises from the KV cache maintained during decoding, whose memory footprint scales with the number of visual tokens in the context. For VLMs based AR assistant, user queries frequently depend on longer temporal context, requiring retrieval over extended video segments. Such long range retrieval further increases memory footprint and inference time, making on device deployment particularly challenging.

Most existing methods either sample temporal frames, such as VidF4 [35] and GenS [62], or select query-relevant spatial regions conditioned on the user's

question, such as DynTok [69] and Q Frame [70]. However, these approaches require the user query as input, so the video must be reprocessed for each question, increasing computation and response latency. In contrast, UTOPI-AR performs pre-compression before the query stage, enabling compressed visual tokens to be stored in the KV cache and reused across queries, which substantially reduces response time. Some approaches, such as DTD [61] and PruneVid [28], also perform temporal or spatial compression without relying on queries. However, they are not tailored to egocentric AR video. These methods mainly target redundancy in static scenes and struggle with the strong redundancy caused by continuous viewpoint translation and rotation common in AR settings. In contrast, UTOPI-AR leverages user motion signals to enable more effective viewpoint-shift-invariant token compression.

Prior work shows that user gaze is highly informative for modeling behavior and intent [32, 34, 47, 50]. However, these studies focus on improving accuracy rather than efficiency, and several [34, 47] introduce heavy pipelines such as optical flow that increase computational overhead. Other approaches [14, 53, 66] use gaze to accelerate VLMs, but typically rely on simple zero-shot heuristics such as central cropping or foveated tokenization without task-specific finetuning. In contrast, UTOPI-AR leverages user gaze signals and bridges gaze fixations with LVU tasks by finetuning a lightweight saliency gating head on subsets of EgoEye and EgoSchema, enabling adaptive refinement of gaze-guided regions of interest for more accurate token compression.

2.2 Attention-Guided Question Formation in Egocentric AR

Human perception is not uniform across the visual field but instead prioritizes attended regions. Classic studies characterize visual attention as a "spotlight" [21, 52], often approximated by gaze position. Attention typically declines with increasing distance from this focus and can be modeled using functions such as a two-dimensional Gaussian [30], resulting in higher fidelity near the attended location and progressively lower fidelity in the periphery [13, 20].

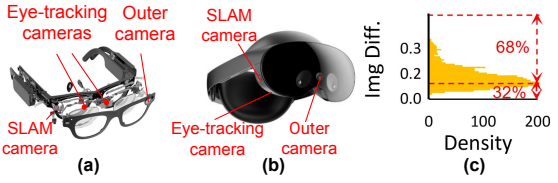


Fig. 2: (a) Meta Aria glasses equipped with multiple cameras. (b) Meta Quest Pro headset, which functions as both an AR and VR device. (c) Distribution of image differences.

This selectivity has direct implications for the types of questions an AR user is likely to ask. As illustrated in Figure 1 (b), a user wearing an AR HMD in a kitchen may fixate on a cup on the counter; objects that receive sustained attentional focus naturally become high-priority candidates for future queries, such as "Where did I leave the cup?". In contrast, nearby objects with weaker attentional signals, shown with lighter bounding boxes in Figure 1 (b), are less likely to be referenced. From a modeling perspective, gaze thus provides a **useful prior for identifying semantically important tokens in egocentric visual**

streams. Consistent with this view, prior work in cognitive psychology and vision science shows that gaze patterns are predictive of subsequent memory formation and information-seeking behavior [33, 64].

2.3 Human Attention in AR Environment

To study AR users' attention patterns over time, we use the *ARIA Everyday* (AEA) dataset [38], collected with Meta Aria smart glasses. The dataset includes over 7.5 hours of egocentric multimodal recordings from more than 150 participants performing everyday activities, with synchronized first-person video, gaze trajectories, and head motion. As shown in Figure 2(c), we measure RGB pixel-wise differences between consecutive frames to quantify visual changes. About 32% of frame pairs show less than 9% change, indicating substantial temporal redundancy and strong correlation across adjacent frames. This suggests that much of the visual content in egocentric AR video remains stable over short intervals, enabling opportunities for efficient processing and computation reuse.

2.4 AR Hardware System

Figure 2 illustrates a representative AR hardware configuration. Typical headsets integrate multiple front and side facing cameras that capture both the surrounding environment and aspects of user state, including gaze direction. Outward facing cameras produce high resolution views of the scene (for example, 1408×1408 on the Meta Aria Glass [2]), while inward facing cameras record lower resolution monochrome images of the eyes. Together, these sensing channels support gaze estimation [1, 2, 46] using either analytic models such as pupil–corneal reflection or modern learning based approaches [37], providing essential signals of user attention and interaction for many AR applications.

In addition, tracking cameras fused with inertial measurements from the onboard Inertial Measurement Unit (IMU) enable robust visual–inertial odometry and full 6-DoF head pose estimation, following pipelines similar to ORB-SLAM3 [12]. IMU and depth sensors are standard in modern AR systems, including Microsoft HoloLens 2 [18], Apple Vision Pro [8], and Magic Leap 2 [41]. Together, this sensing stack provides high-quality visual input, gaze information, and precise pose tracking at low cost, making AR platforms well suited for fine-grained, intention-aware segmentation and interaction.

Mainstream AR headsets typically provide only 8-16 GB of on device memory [7, 45]. Many of them adopt a unified memory architecture [7, 41, 45], where the CPU and GPU share same memory pool. In practice, the system also reserve part of this memory, leaving an even smaller budget for model deployment. Meanwhile, their compute is often constrained by mobile or low power chips. This often leads to high response latency for multimodal models. Together, these memory and compute constraints make deploying LVU models on AR challenging.

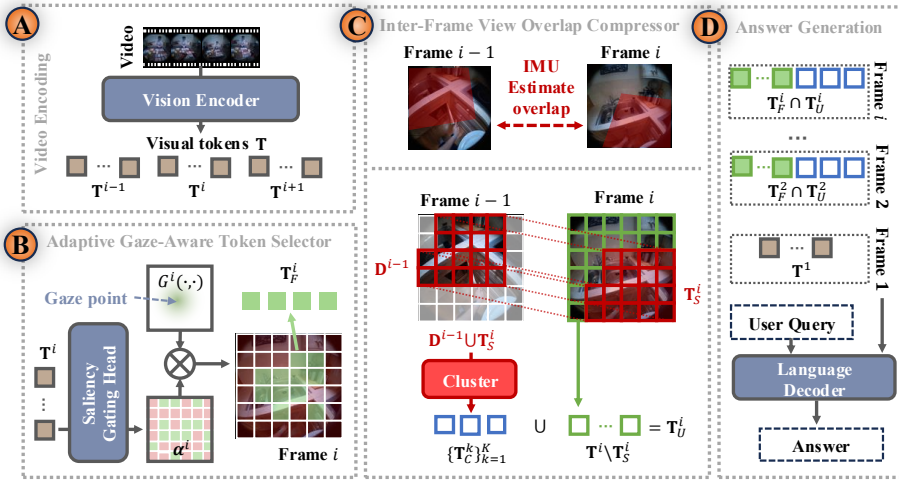


Fig. 3: The overall architecture of UTOPI-AR.

3 Methodology

3.1 UTOPI-AR Overview

VLMs process a video by first sampling frames to form a frame set $\mathbf{I}$, and then encoding each frame using a ViT-based vision encoder to produce visual tokens $\mathbf{T} = \{\mathbf{T}^i\}_{i=1}^N$, where $\mathbf{T}^i = \{\mathbf{T}^i_{w,h} \mid w \in [1, W],\ h \in [1, H]\}$, where i denotes the temporal index of frame $\mathbf{I}^i$, N is the number of sampled frames, and (w, h) represents the spatial location of token $\mathbf{T}^i_{w,h}$ within frame $\mathbf{I}^i$, with an $H \times W$ token grid. Positional embeddings $P(\cdot)$ are then applied to these tokens to obtain the embedded representation $\mathbf{T}_{\text{emb}} = P(\mathbf{T})$. Finally, $\mathbf{T}_{\text{emb}}$ is fed together with textual tokens into the language decoder to generate the final output. To reduce computational and memory overhead on AR devices, our goal is to compress the number of visual tokens in $\mathbf{T}_{\text{emb}}$, and thereby reducing the size of the key–value (KV) cache required for subsequent user queries.

The overall UTOPI-AR pipeline, shown in Figure 3, consists of four stages: *Video Encoding*, *Adaptive Gaze-Aware Token Selector (AGTS)*, *Inter-Frame View Overlap Compressor (IVOC)*, and *Answer Generation*. After obtaining visual tokens $\mathbf{T}$ for all frames from the video encoding stage, we first apply the AGTS to each frame $\mathbf{I}^i$. The AGTS leverages real-time gaze information together with a fine-tuned saliency gating head to assign a saliency score to each visual token, and selects a subset of foreground tokens $\mathbf{T}^i_F \subset \mathbf{T}^i$ that are highly salient and closely related to the user's activity. We then apply IVOC to further eliminate temporal redundancy within $\mathbf{T}^i$, producing a reduced token set $\mathbf{T}^i_U$. The final compressed visual token set is given by $\mathbf{T}' = \mathbf{T}^1 \cup \{P(\mathbf{T}^i_F) \cap P(\mathbf{T}^i_U)\}_{i=2}^N$ Next, we describe the implementation details of the AGTS and IVOC modules.

3.2 Adaptive Gaze-Aware Token Selector (AGTS)

According to Section 2.2, gaze location provides a strong prior for identifying semantically important tokens in egocentric visual systems. As discussed in Section 2.4, modern AR devices are equipped with eye-tracking modules that record the 2D gaze coordinates (g_x^i, g_y^i) for each frame $\mathbf{I}^i$. We leverage this signal by applying a 2D Gaussian kernel centered at (g_x^i, g_y^i) on frame i:

$$G^i(w,h) = \exp\left(-\frac{(w-g_x^i)^2 + (h-g_y^i)^2}{2\sigma^2}\right), \tag{1}$$

where σ is a predefined standard deviation. To further refine token selection for question answering, we apply a fine-tuned two-layer MLP saliency gating head to each visual token $\mathbf{T}_{w,h}^i \in \mathbf{T}^i$ to predict a gate value $\alpha_{w,h}^i = \mathrm{MLP}(\mathbf{T}_{w,h}^i)$. The final foreground tokens are then selected as:

$$\mathbf{T}_F^i = \{\mathbf{T}_{w,h}^i \mid \alpha_{w,h}^i \cdot G^i(w,h) > \rho\}. \tag{2}$$

Here, ρ is a predefined threshold, and w, h denote the spatial location of token $\mathbf{T}_{w,h}^i$. The positional indices (i, w, h) of each foreground token are kept identical to the original, preserving spatiotemporal structure of foreground tokens.

To better align the user gaze distribution with the LVU task, we fine-tune the MLP using a calibration dataset while freezing the remaining components of the VLM to reduce training cost. Specifically, during the forward pass, after obtaining $\mathbf{T}_F^i$ via Equation 2, we compute the final loss using next-token prediction $\mathcal{L}_{NTP} = -\log \frac{\exp(\mathbf{z}_y)}{\sum_{v=1}^{V}\exp(\mathbf{z}_v)}$, where $\mathbf{z} = F_{Dec}(\mathbf{T}_F^i)$. $F_{\mathrm{dec}}(\cdot)$ denotes the decoder function, $\mathbf{z} \in \mathbb{R}^V$ represents the predicted logits over the token vocabulary, where V is the vocabulary size, and $\mathbf{z}y$ is the logit corresponding to the correct answer token. Since the language decoder is differentiable, we can compute $\frac{\partial \mathcal{L}\mathrm{NTP}}{\partial \mathbf{T}_F^i}$.

However, $\mathbf{T}_F^i$ is obtained from $\mathbf{T}_{w,h}^i$ via the hard selection defined in Equation 2 and is therefore not a differentiable function of the gate values $\boldsymbol{\alpha}^i$. As a result, $\frac{\partial \mathbf{T}_F^i}{\partial \boldsymbol{\alpha}^i}$ is undefined. We thus adopt a straight-through estimator (STE) and approximate the gradient with respect to $\alpha_{w,h}^i$ as follows:

$$\frac{\partial \mathbf{T}_{w,h}^i}{\partial \alpha_{w,h}^i} \approx G^i(w,h) \sum_{d=1}^{D} \mathbf{T}_{w,h,d}^i, \forall\ \mathbf{T}_{w,h}^i \in \mathbf{T}_F^i \tag{3}$$

where $\mathbf{T}_{w,h}^i \in \mathbb{R}^D$ is a D-dimensional token embedding vector. Since $\boldsymbol{\alpha}^i$ is produced by saliency gating head, we can backpropagate the gradients to the saliency gating head using Equation 3.

3.3 Inter-Frame View Overlap Compressor

Because egocentric videos are typically captured as a continuous first-person stream, they exhibit strong scene continuity over time, as illustrated in Fig. 4.

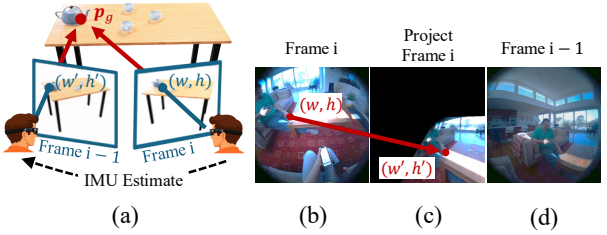


Fig. 4: IMU based inter-frame coordinate projection.

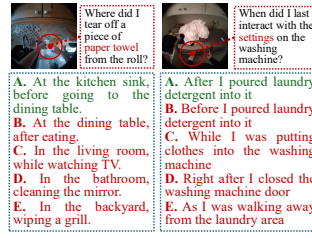


Fig. 5: EgoEye QA example.

Content observed by the user persists across frames, while the corresponding tokens often shift spatially due to changes in head pose and viewpoint. For instance, Frame i and Frame $i-1$ in Fig. 4 illustrate two consecutive frames from the AEA dataset at times $i-1$ and i, where the man and table are visible in both frames. We observe that corresponding visual tokens generally reappear at different spatial locations rather than remaining aligned at the same position.

In addition, as described in Section 2.4, tracking cameras combined with measurements from the onboard IMU enable robust visual–inertial odometry and accurate 6-DoF head pose estimation. IMU sensors are a standard component in modern AR systems [8, 18, 41]. Consequently, AR devices continuously record the user's head pose, including both position and orientation, during operation.

Although consecutive frames exhibit high temporal redundancy, directly comparing their RGB values often produces large differences. This is primarily caused by changes in the user's head pose, which alter the camera's orientation and position between frames. As a result, the same scene content may appear at different spatial locations in the image. Therefore, user pose information must be considered when evaluating redundancy across frames.

To achieve this, we adopt a geometry-based reprojection strategy to estimate inter-frame overlaps. Specifically, for each token $\mathbf{T}^i_{w,h}$ in the current frame $\mathbf{I}^i$, its camera coordinate is denoted as (w, h). Using the camera intrinsics, this coordinate can be mapped to a global coordinate $\mathbf{p}_g = \text{proj}_{c\to g}(w, h) \in \mathbb{R}^3$. Next, based on the camera pose change estimated from the IMU, we recover the camera pose of the previous frame $\mathbf{I}^{i-1}$ and reproject the global coordinate $\mathbf{p}_g$ back to that frame. This yields the corresponding camera coordinate $(w', h') = \text{proj}_{g\to c}(\mathbf{p}_g)$ in $\mathbf{I}^{i-1}$. We denote the mapping of camera coordinates between two frames as $f(\cdot, \cdot)$, such that $f(w, h) = (w', h')$. If (w', h') falls within the field of view of $\mathbf{I}^{i-1}$, then we select the token $\mathbf{T}^{i-1}_{f(w,h)}$ in the $\mathbf{I}^{i-1}$ whose camera coordinate matches (w', h'). The regions represented by $\mathbf{T}^{i-1}_{f(w,h)}$ and $\mathbf{T}^i_{w,h}$ therefore correspond to the **overlapping area** between two frames, indicated by the top part of Figure 3 (c). The detailed mapping formulas are provided in the Appendix.

To eliminate the temporal redundancy, for the overlapped visual regions, we keep only one full copy in the earlier frame and replace the corresponding content in later frames with compressed **surrogate tokens**. The process is highlighted in the bottom portion of Figure 3 (c). First, we identify the inter-frame overlap region $\mathbf{O}^i$ between frames $\mathbf{I}^i$ and $\mathbf{I}^{i-1}$ and collect the corresponding tokens $\mathbf{T}^i_O =$

$\{\mathbf{T}^i_{w,h} | (w,h) \in \mathbf{O}^i\}$. We then use $f(\cdot,\cdot)$ to map coordinates of $\mathbf{T}^i_O$ into $\mathbf{I}^{i-1}$ and compute the cosine similarity with the tokens at the corresponding locations:

$$s^i_{w,h} = \text{cosim}\big(\mathbf{T}^{i-1}_{f(w,h)},\ \mathbf{T}^i_{w,h}\big), \tag{4}$$

where $\text{cosim}(\cdot,\cdot)$ denotes the cosine similarity. We then identify tokens within $\mathbf{O}^i$ whose representations remain sufficiently similar to those in $\mathbf{I}^{i-1}$:

$$\mathbf{T}^i_S = \big\{\mathbf{T}^i_{w,h} \in \mathbf{T}^i_O \mid s^i_{w,h} > \theta\big\}, \tag{5}$$

where θ is the similarity threshold. Next, We find counterparts of $\mathbf{T}^i_S$ in the previous frame $\mathbf{I}^{i-1}$,

$$\mathbf{D}^{i-1} = \big\{\mathbf{T}^{i-1}_{f(w,h)} \mid s^i_{w,h} > \theta,\ (w,h) \in \mathbf{O}^i\big\}, \tag{6}$$

and cluster these tokens $\mathbf{D}^{i-1} \cup \mathbf{T}^i_S$ to obtain a compressed representation. This produces K cluster centers $\{\mathbf{T}^k_C\}_{k=1}^K$, which are used as surrogate tokens to replace $\mathbf{T}^i_S$, thereby significantly reducing the total number of tokens. During subsequent VLM processing, for each surrogate token $\mathbf{T}^k_C$, we assign its temporal position to the current frame index i, and set its spatial position to the mean spatial coordinate of all tokens in the corresponding cluster. This coordinate is then mapped back to $\mathbf{I}^i$ via the inverse mapping $f^{-1}(\cdot,\cdot)$ and used as the final spatial location.

3.4 EgoEye Collection

While multiple benchmarks have been proposed for LVU, they primarily focus on generic video-based queries and do not fully capture the **human-centric** and **attention-guided** nature of real AR usage, where questions are closely tied to user attention patterns [11, 26, 63]. Motivated by this gap, and building on the background presented in Section 2.3, we introduce a new benchmark, termed *EgoEye*, for LVU that simulates realistic, attention-driven interactions in AR environments.

EgoEye is constructed from real-world egocentric video datasets with authentic gaze tracking, including Aria Everyday Activities (AEA) [38] and Nymeria [39]. In contrast to prior LVU benchmarks, our questions are explicitly **gaze-related**: we select question subjects by leveraging gaze traces to identify attended regions and objects, and then sample question targets with a strong bias toward items spatially close to the user's fixation points over time. This design ensures that the queries reflect what the user actually looked at and interacted with during the experience. To scale the creation of questions, we employ AI-assisted generation to produce multiple-choice questions, followed by extensive human review and rule-based filtering to ensure quality and consistency. EgoEye includes video clips ranging from 50 seconds to 5 minutes, with an average length of 2 minutes and 31 seconds per clip. We extract more than 2k question subjects around users' gaze locations, resulting in over 5k QA pairs. Fig. 5 provides some QA examples for reference, with their question subjects being "Paper Tower" and "Washing Machine Knob", respectively. Detailed statistics and more QA examples are provided in the Appendix.

4 Experiment

4.1 Evaluation Setups and Implementation Details

Video-QA Benchmarks To evaluate UTOPI-AR, we use three Video Question–Answering (VQA) benchmarks: EgoSchema [43], EgoGazeVQA [51], and our proposed EgoEye. As UTOPI-AR targets AR scenarios, all selected datasets are based on egocentric videos and multiple-choice questions. EgoSchema provides over 5k automatically constructed five-choice questions and offers a broad test of long-context understanding, reasoning, and retrieval. EgoGazeVQA supplies user gaze traces, with questions categorized into causal, spatial, and temporal reasoning. Both benchmarks are derived from the egocentric video dataset Ego4D [25]. Since UTOPI-AR requires signals such as user gaze and IMU measurements, we obtain the original Ego4D metadata and align it to recover the necessary signals. For videos missing gaze signals, we estimate user gaze using [27], and further include an ablation study on the impact of gaze noise in Table 6. Our EgoEye contains over 5k multiple-choice QA pairs (more detail in the Appendix) and includes user gaze and IMU metadata. In contrast to existing benchmarks, our questions are explicitly centered on human attention signals and realistic AR usage, producing queries that better reflect what users actually look at and care about, and enabling a more natural evaluation setting for long-context egocentric video understanding in AR environments.

Implementation Details For video input, we set the frame resolution to 532×532 and uniformly sample 300 frames per video for experiments on EgoEye and EgoSchema. For EgoGazeVQA, we sample 100 frames per video, prioritizing key frames that contain gaze points. To ensure a consistent token budget across all experiments for fair comparison, we adaptively adjust hyperparameters θ and ρ so that each video retains exactly the target ratio of visual tokens. We test the algorithms on the lightweight models including Qwen2.5VL-7B [10], Videollama3-7B [67], PLLaVA [60] and InternVL2.5-8B [15] and use four NVIDIA A6000 (48GB) GPUs for all accuracy evaluation. Finally, we set the number of cluster centers K to 2% of the total number of tokens in the cluster, i.e., $K = 0.02 \times |\mathbf{T}_S^i|$. To evaluate UTOPI-AR, we compare it against other methods for token pruning of LVU task, including PruneVid [28], DTD [61] and MindPalace [29]. We also evaluate *Random Drop* where the token are dropped randomly. For PruneVid [28], we disable its KV-cache compression module because it requires modifications to Qwen2.5VL and some implementation details are not publicly available. For MindPalace [29], we adopt Qwen2.5VL as the backbone LLM for generating final answers. When fine-tuning the saliency gating head, we randomly sample 1k QA pairs from EgoEye as the training set, using the remaining data for evaluation. During training, we interpret the gate values as probabilities for stochastic token dropping. We start with a low drop ratio and gradually increase it, while simultaneously annealing the temperature.

4.2 Performance on Benchmarks

In Table 1, we evaluate several VLM baselines and combinations of Qwen2.5VL with different compression methods on three benchmarks: EgoEye (**EE**), EgoSchema (**ES**), and EgoGazeVQA (**EGV**). All methods are evaluated under identical token budgets. During evaluation, we use the same prompt template for all models and provide the video, question, and answer options in a fixed order. The *Token Budget* in the table denotes the ratio of the number of visual tokens after compression to that before compression. A token budget marked with "*" indicates a value of 0%, meaning that only the question and answer options are provided to the model without any video input; this setting corresponds to a *blind test*. Some token budgets are marked with "†" to indicate that these models do not apply token compression; instead, we downsample the video resolution to ensure that the total number of visual tokens matches the specified budget.

Table 1: Acc. Evaluation of UTOPI-AR.

Method	Token Budget	EE	ES	EGV
QWen2.5VL-7B	100%	58.4%	60.1%	47.8%
QWen2.5VL-7B	0%*	36.6%	49.4%	43.4%
QWen2.5VL-7B				
w/ UTOPI-AR (ours)	15%	**60.6%**	**64.8%**	**49.0%**
w/ DTD [61]	15%	57.6%	64.5%	47.9%
w/ PruneVid [28]	15%	51.3%	63.0%	48.8%
w/ Random Drop	15%	56.8%	59.3%	46.1%
QWen2.5VL-7B	15%†	50.4%	53.8%	44.2%
MindPalace [29]	15%†	33.2%	45.2%	37.9%
PLLaVA [60]	15%†	34.1%	29.6%	31.5%
Videollama3-7B [67]	15%†	46.4%	49.7%	42.5%
InternVL2.5-8B [15]	15%†	52.4%	51.5%	43.0%
QWen2.5VL-7B				
w/ UTOPI-AR (ours)	50%	**57.1%**	**61.9%**	**48.5%**
w/ DTD	50%	57.0%	60.2%	47.2%
w/ PruneVid	50%	52.4%	60.4%	47.8%
w/ Random Drop	50%	56.4%	59.6%	46.9%
w/ UTOPI-AR (ours)	30%	**57.4%**	**62.6%**	**48.7%**
w/ DTD	30%	56.6%	62.1%	48.2%
w/ PruneVid	30%	53.3%	61.9%	48.1%
w/ Random Drop	30%	56.1%	60.3%	45.8%

Experimental results show that under the three token budgets $(15\%, 30\%, 50\%)$, our proposed UTOPI-AR method consistently outperforms all other efficiency baselines across all benchmarks. Specifically, when applied to Qwen2.5VL, UTOPI-AR retains only 15% of the visual tokens while still improving accuracy by an average of 2.7% compared to using the full set of visual tokens. Compared with other compression methods, UTOPI-AR achieves at least 2.5% higher accuracy on average. In addition, relative to naive uniform-resolution downsampling of Qwen2.5VL, UTOPI-AR improves the average accuracy by 8.6%. Finally, compared with the blind test setting, UTOPI-AR uses only 15% of the visual tokens yet achieves an average accuracy gain of 15%, further demonstrating the effectiveness of our approach.

4.3 UTOPI-AR System Performance

Latency and Memory To evaluate the system performance of UTOPI-AR, we implement the system on both a desktop GPU (RTX A6000) and an edge GPU (NVIDIA Jetson Orin NX), and report latency and peak memory usage. These metrics capture the responsiveness and resource efficiency of the system, which are critical for real-time AR workloads deployed on resource-constrained edge devices. NVIDIA Jetson Orin NX has been widely used in prior work [3, 4, 49] to simulate AR/VR devices. This platform provides 16GB of unified memory. Since the VLMs evaluated in Table 1 cannot fully fit within the memory capacity of

Table 2: Latency and Peak Memory Evaluation. "w/ ours" indicates that UTOPI-AR is implemented over the corresponding model.

Method	platform	Token budget	Peak Memory Cost (w/o Model weights)			UTOPI-AR Overhead		TTFT (speed up×)	TPS (speed up×)	EE Acc.
			VE	LD	Overall (save%)	MFLOPs	Latency			
QWen 7B	GPU	100%	6.6G	19.8G	19.8G	–	–	19.3s	7.5	58.4%
QWen 7B w/ ours	GPU	15%	6.6G	4.1G	6.6G (**67%**)	1636	673ms	8.7s (**2.2**×)	26.6 (**3.5**×)	60.6%
QWen 7B (4bit)	Jetson Orin	100%	3.2G	7.1G	**OOM** 7.1G	–	–	**OOM** 173s	**OOM** 0.43	57.3%
QWen 7B (4bit) w/ ours	Jetson Orin	15%	3.2G	1.8G	3.2G (**55%**)	545	980ms	88s (**2.0**×)	1.3 (**3.0**×)	57.5%
QWen 3B (4bit)	Jetson Orin	100%	2.9G	6.4G	6.4G	–	–	122s	0.73	52.0%
QWen 3B (4bit) w/ ours	Jetson Orin	25%	2.9G	2.2G	2.9G (**55%**)	545	980ms	65s (**1.9**×)	2.0 (**2.7**×)	52.1%
SmolVLM 2.2B	Jetson Orin	100%	2.0G	2.9G	2.9G	–	–	21s	3.2	46.3%
SmolVLM 2.2B w/ ours	Jetson Orin	60%	2.0G	1.8G	2.0G (**32%**)	360	440ms	13s (**1.6**×)	4.8 (**1.5**×)	45.9%

the edge GPU, we adopt 4-bit quantized Qwen2.5VL-7B [58] and Qwen2.5VL-3B [58] as baselines, using publicly available checkpoints from Hugging Face. We also evaluate UTOPI-AR on SmolVLM [44].

Table 2 reports the *Peak Memory Cost*, defined as the maximum memory usage during inference. We exclude the static memory occupied by model weights and report only the dynamic components, such as layer activations and the KV cache. **VE** denotes the peak memory cost of the Vision Encoder, while **LD** denotes that of the Language Decoder. **Overall** corresponds to the maximum of the two (highlighted in yellow). We observe that without compression, LD becomes the primary bottleneck due to the large KV cache, with memory usage on average 2.2× higher than that of the VE. With UTOPI-AR, while maintaining comparable accuracy, the peak memory consumption is reduced by an average of 52.2%. In particular, Qwen2.5VL-7B (4-bit) runs out of memory without the introduction UTOPI-AR (the red entry in the corresponding row represents a reasonable estimate).

Additionally, we report latency results for UTOPI-AR in terms of *Time To First Token* (TTFT), which measures the time from input to the first generated token, and throughput, i.e., Tokens Per Second (TPS), in Table 2. Similar metrics have been used in prior work to evaluate the system performance of large models [5, 24, 65]. The results show that both response latency and decoding throughput improve substantially, achieving average gains of 1.92× and 2.67×, respectively. Overall, these results demonstrate that UTOPI-AR significantly reduces memory and latency bottlenecks while preserving accuracy, making large VLMs more practical for deployment in AR environments.

Performance Under Different Budget: To evaluate the trade-off between accuracy and token budget, we apply UTOPI-AR to PLLaVA [60] and Qwen2.5VL-7B and compare results under different token budgets on two benchmarks (Table 3). A token budget of 100% denotes the original model. The best accuracy is highlighted in bold, and the smallest token budget that matches or exceeds the baseline accuracy is underlined. Results show that compressing visual tokens with UTOPI-AR often improves accuracy on EgoSchema, likely because removing redundant tokens reduces interference. With Qwen2.5VL, our method compresses up to 95% of visual tokens without accuracy loss. On PLLaVA (EgoSchema), UTOPI-AR achieves up to 85% token reduction while maintain-

Table 4: Evaluation of different AGTS / IVOC configurations

AGTS	IVOC	Token Budget	ES
✓	✓	15%	64.8%
✓	×	50%	64.1%
×	✓	30%	64.4%
×	×	100%	60.1%

Table 5: Performance comparison on the Low-Motion subset "SLOW" and Motion-Intensive subset "FAST".

Method	Token Budget	ES		
		FAST	SLOW	AVG
QWen w/ ours	30%	62.0%	63.2%	62.6%
QWen w/ ours	15%	63.6%	66.0%	64.8%
QWen w/ ours	5%	59.2%	63.6%	61.4%

Table 6: Impact of gaze noise.

Gaze Noise	EE	EGV
0 px	60.6%	49.0%
75 px	60.3%	48.7%
150 px	60.4%	47.7%
225 px	58.9%	47.4%
300 px	57.7%	47.1%

ing performance. On EgoEye, accuracy drops by less than 2% at a 15% token budget. Overall, UTOPI-AR achieves a strong balance between efficiency and accuracy across models and datasets.

Impact of AGTS and IVOC To validate the effectiveness of the AGTS and IVOC modules, we conduct ablation studies, as shown in Table 4. We evaluate Qwen2.5VL-7B with UTOPI-AR under different configurations on EgoSchema. The results show that enabling either AGTS or IVOC improves accuracy to around 64%. When both are enabled, accuracy reaches 64.8% while achieving the lowest token budget. Since AGTS and IVOC address spatial and temporal redundancies, respectively, they complement each other and jointly improve compression efficiency without harming accuracy, enabling both lower token budgets and better overall performance.

Table 3: Budget Analysis.

Baseline	Token Budget	EE	ES
PLLaVA	100%	**34.8%**	37.8%
w/ UTOPI-AR	50%	33.7%	39.8%
w/ UTOPI-AR	30%	34.2%	**40.1%**
w/ UTOPI-AR	15%	33.5%	39.3%
w/ UTOPI-AR	5%	31.8%	32.1%
QWen2.5VL-7B	100%	58.4%	60.1%
w/ UTOPI-AR	50%	57.1%	61.9%
w/ UTOPI-AR	30%	57.4%	62.6%
w/ UTOPI-AR	15%	**60.6%**	**64.8%**
w/ UTOPI-AR	5%	53.4%	61.4%
w/ UTOPI-AR	3%	50.5%	59.4%

Impact of Object-Gaze Distance To evaluate the robustness of the AGTS, we group questions in the EgoEye benchmark by the distance between the question subject and the gaze point. We then compare the accuracy of Qwen2.5VL-7B with different compression methods across these distance groups (Figure 6). Note that EgoEye frames have a resolution of 1408×1408. As the distance increases, all models exhibit lower accuracy, likely because objects far from the gaze point tend to appear near the frame boundary, where visual cues are less stable and questions become more challenging. When the question subject is close to the gaze point, UTOPI-AR achieves about 5% higher accuracy, demonstrating the benefit of AGTS in leveraging user attention to preserve informative regions. Even when the subject is far from the gaze point, UTOPI-AR maintains comparable accuracy to other methods, indicating strong robustness.

Impact of Motion-Intensive Video To assess the robustness of the IVOC module, we compare QWen2.5VL-7B with UTOPI-AR on *Motion-Intensive* and *Low-Motion* videos (see Table 5). To construct the evaluation split, we first encode each EgoSchema video with the QWen2.5VL vision encoder and obtain a token map for every frame, denoted $\mathbf{T}^i$. We then average all token embeddings within each frame to get a single frame-level representation, $\mathbf{r}_i = \text{mean}(\mathbf{T}^i)$.

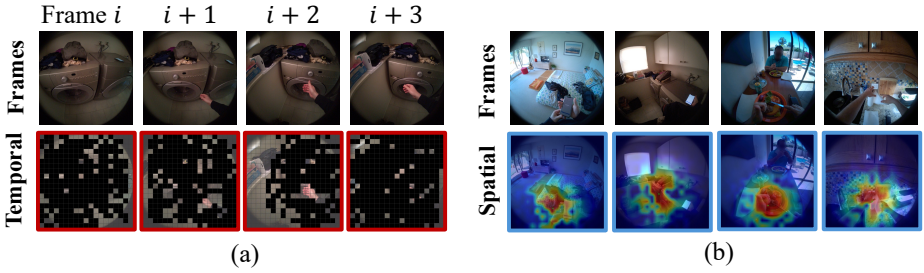
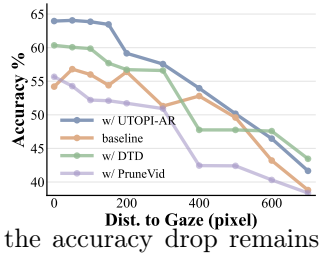


Fig. 7: Visualization of AGTS and IVOC.

Next, we compute the cosine similarity between consecutive frame representations and average over the video to obtain the mean inter-frame similarity. $\alpha = \frac{1}{N}\sum_{i=1}^{N} \text{cosim}(\mathbf{r}_i, \mathbf{r}_{i+1})$, We sort all EgoSchema videos by α and select the top 50% (larger α) as *Low-Motion* videos (denoted as SLOW); the remaining 50% are *Motion-Intensive* videos (denoted as FAST). As shown in Table 5, under the same token budget, SLOW videos generally achieve higher accuracy than FAST ones. However, the gap is small, averaging 2.6%, indicating that IVOC remains robust across videos with different levels of temporal redundancy.

Impact of Gaze Noise To evaluate the impact of gaze-signal noise on UTOPI-AR, we inject perturbations of varying magnitudes into the gaze signals in EgoEye and EgoGazeVQA, ranging from 75 pixels to 300 pixels. For instance, at a noise level of 75 pixels, the perturbed gaze point is uniformly sampled within a circle centered at the ground-truth gaze point with a radius of 75 pixels. Table 6 reports the results, where the baseline is Qwen2.5-VL-7B with a token budget of 15%. All other settings follow Table 1. Even under large gaze noise, the accuracy drop remains within 3%. These results show the robustness of our method.

Fig. 6: Impact of different "Question-Gaze" distance.



Visualization We provide several visualization clips in Fig. 7 for reference. (a) shows the IVOC toke drop, where the black mask indicates dropped tokens. (b) shows the AGTS saliency heat map of corresponding frame.

5 Conclusion

We propose UTOPI-AR, a plug-and-play user behavior guided token pre compress module in AR environments. UTOPI-AR improves efficiency by using gaze-informed attention to discard spatially redundant visual tokens. Experiments show that UTOPI-AR compresses up to 95% of visual tokens on Qwen2.5VL and 85% on PLLaVA while maintaining baseline accuracy. It achieves the highest accuracy (64.8%) under the strongest compression and remains robust across motion levels, with only a 2.6% gap. Our method provides an effective solution for improving long-video understanding efficiency in AR/VR scenarios.

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